

1.) HTML

→ HTML is a Standard Markup Language Used to Create and Structure webpages.

→ HTML tell the browser what the content is and how it should be structured.

HTML → Stands for Hypertext Markup language.

2.) HTML is called as Markup language because it uses tags to mark and describe the structure of content.

Ex: <h1> my heading </h1>

<p> my paragraph </p>

⇒ HTML is not a programming language because it does not contain programming logic such as conditions, loops or calculations. It is a markup language.

3.)

HTML TAG	HTML ELEMENT
→ HTML tag is a keyword enclosed inside angle brackets <code><></code>. → Tags tell the browser what type of content it is dealing with Ex: <code><p></code> → Represents a part of the Structure	HTML Element is the complete structure consisting of the opening tag, content and closing tag. Ex: <code><p> Hello </p></code> → Represents the complete HTML Structure

4.) `<!DOCTYPE html>`:

- `<!DOCTYPE html>` tells the browser that the document uses HTML 5.
- It is not an HTML tag. It is a DOCTYPE declaration.

5.) Basic Structure of an HTML Document:

```

<!DOCTYPE html>
<html>
  <head>
    <title> my page </title>
  </head>
  <body>
    <h1> My page </h1>
    <p> welcome to my website </p>
  </body>
</html>

```

PURPOSES:

`<html>` → Root Element of the HTML document.

`<head>` → Contains information about the page, such as title, metadata, CSS.

`<body>` → Contains the visible content of the webpage.

6.) HTML ATTRIBUTES:

→ An attribute provides additional information about an HTML element.

→ Attributes are written inside the opening tag.

Syntax:

`<tag attribute = "value"> content`

Ex:

`<p class = "Para"> Hello Students </p>`

Here,

`<p>` → Tag

`class` → attribute

`"Para"` → attribute value

`Hello Student` → content

`</p>` → closing Tag.

7.) Between Src & href.

Src	href
→ Src Means Source → It Specifies the location of a resource that is Embedded or Loaded into the page. Ex: <pre></pre>	→ href stands for hypertext reference → It is commonly used with the <a> Element. Ex: <pre> google </pre>

8.) PURPOSE OF THE alt ATTRIBUTE IN AN TAG

The alt attribute provides alternative text for an image.

Ex:

```
<img Src = "Profile.jpg" alt = "Student Profile Photo">
```

9.) BETWEEN BLOCK-LEVEL AND INLINE ELEMENTS:

BLOCK-LEVEL ELEMENTS	INLINE ELEMENTS
→ Starts on a new line → Generally takes available width → Can contain other elements depending on HTML rules.	→ Does not normally start on a new line → Takes only required width → Usually used for small pieces of content.

→ EX: <div>

→ EX: <p>

→ EX: <Section>

→ EX:

→ EX: <a>

→ EX:

10) SEMANTIC HTML:

Semantic HTML uses elements that clearly describe the meaning and purpose of their content.

Ex: <header> Header </header>

<nav> navigation </nav>

<main> main content </main>

<section> section </section>

<footer> footer </footer>

BETWEEN SEMANTIC AND NON-SEMANTIC ELEMENTS:

SEMANTIC ELEMENTS	NON-SEMANTIC ELEMENTS
→ Has Meaningful name → Clearly describes Purpose → Better page Structure → Helps accessibility → <header> → <nav> → <article> → <section> → <footer>	→ Generic Name → Does not describe Purpose → Mainly used as containers. → Less Meaningful by itself. → <div> →

11.) BETWEEN `<div>` & `<span>`

<code><div></code>	<code></code>
→ <code><div></code> is a generic block-level container <pre><div> <h2> Student details </h2> <p> name: Max </p> </div></pre>	→ <code></code> is a generic inline container. <pre><p> my name is Max </p></pre>
→ It is non semantic because the name <code><div></code> doesn't describe the meaning of its content	→ It is also non-semantic.
→ used mainly for 1. Grouping Elements 2. layout 3. Styling 4. JavaScript targeting	→ used mainly for. 1. Grouping inline content 2. Styling part of text 3. JavaScript targeting.

12.) Different Types of HTML Lists

There are three main type of lists.

1. Unordered List `<ul>` → Display items with bullets
2. Ordered List `<ol>` → Display items in a numbered order.
3. Description List `<dl>` → used for terms and their descriptions.

1. Unordered list

 HTML

 CSS

 JavaScript

Output

- HTML
- CSS
- JavaScript

2. Ordered list

 learn HTML

Output

1. Learn HTML

3. Description list

<dl>

<dt> HTML </dt>

<dd> Structures webpage. </dd>

<dt> CSS </dt>

<dd> Styles webpages. </dd>

</dl>

13. IMPORTANT HTML TABLE TAGS:

1. `<table>` → Create an HTML table

`<table>`

`</table>`

2. `<tr>` → Create a table row.

`<tr>`

`</tr>`

3. `<th>` → Create a table header cell.

`<th> name </th>`

4. `<td>` → Create a table data cell.

`<td> max </td>`

5. `<thead>` → Groups the ~~main body~~ ^{header} rows of a table.

6. `<tbody>` → Groups the ~~header~~ ^{main body} rows of a table

7. `<tfoot>` → Group the footer row of a table.

14.) PURPOSE OF `<meta charset = "UTF-8">`

`<meta charset = "UTF-8">` It specifies the Character Encoding of the webpage.

UTF-8 → Supports many languages and Special Characters.

VIEWPORT META TAG:

→ It makes the webpage display properly on different

Screen Sizes, Especially Mobile devices.

`<meta name="viewport" content="width=device-width,
initial-scale=1.0">`

`width=device-width` → Matches the page width to the device screen.

`initial-scale=1.0` → Sets the initial zoom level to 100%.

15.) OPEN GRAPH (OG):

Open graph (OG) Provides metadata that social platforms can use to create richer previews when a webpage URL is shared.

PURPOSE OF:

1. og:title → Specifies the title displayed when the page is shared.

`<meta property="og:title" content="my website">`

2. og:type → Specifies the type of content

`<meta property="og:type" content="website">`

3. og:image → Specifies the images displayed when the page is shared

`<meta property="og:image" content="https://ex.com/image.jpg">`

4. og: url → Specifies the URL of the webpage.

<meta property="og:url" content="https://example.com">

5. og: description → Provides a short description of the webpages.

<meta property="og:description" content="welcome to my website.">